

Low-Cost Real-time Driver Drowsiness Detection based on Convergence of IR Images and EEG Signals

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Abstract—This paper focused on low-cost real-time driver's drowsiness detection by fusing facial image information obtained through the IR camera (Infrared Camera) and EEG (Electroencephalogram) signal acquired through the EEG sensor. The proposed method was tested on the target board(i.MX6Quad). The i.MX6Quad is the SoCs (System-on-Chip) that integrate many processing units into one die, like the main CPU, a video processing unit and a graphics processing unit for instance. Instead of the RGB camera, the IR camera is applied to driver condition monitoring and drowsiness detection technology by extracting the driver's facial feature information robustly against daytime, night-time, and frequent change of brightness around the face. The headphone type EEG sensor is also used to minimize the user's discomfort. On the target board, the processing time per image frame is about 60ms, so that it can process about 17 frames per second. This processing time can be suitable for the driver's drowsiness detection systems.

Index Terms—face landmark, drowsiness, driver state monitoring, EEG, IR-image

I. INTRODUCTION

The main cause of traffic accidents in recent years is drowsiness driving and attention reduction. To prevent this, driver drowsiness detection and fatigue state recognition technology are widely studied. The driver's condition monitoring technique was mainly based on the driver's facial image information. From the facial image information, features such as facial motion, eye flicker, and pupil state were extracted to predict drowsiness [1]–[3]. Recently, CNN (Convolutional Neural Network) based driver's drowsiness detection methods are proposed, which identifies candidates with RP (Region Proposals) and verifying the candidates based on CNN framework. The CNN based detection method has high detection accuracy [4]–[8], however, it requires a long processing time. It takes several seconds to process one frame in the current driver state monitoring system which does not have high performance GPUs. On the other hand, there is a research to detect the

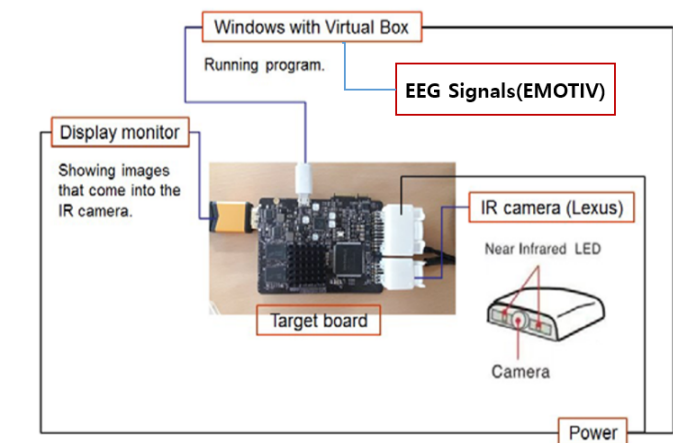


Fig. 1. Image of target board(1GHz i.MX6Quad and 1GB RAM)

drowsiness state by utilizing bio-metric information features such as EEG (Electroencephalogram) through the EEG sensor [3], [9], [10]. The driver condition monitoring technology based on the RGB camera has difficulty in recognizing the driver's condition due to the presence of the face and feature point detection performance deviation depending on the day-time, night-time and frequent brightness changes around the face. EEG based drowsiness detection technology has an important effect on the sleep stage estimation performance by the accurate measurement of the driver's EEG signal through the EEG sensor. However, existing EEG measurement devices have limitations in that it is difficult to accurately measure EEG signals due to inconvenient wearing methods. In this paper, we focused on low-cost real-time driver's drowsiness detection by fusing facial image information obtained through

the IR camera (Infrared Camera) and EEG signal acquired through the EEG sensor. The proposed method tested in the target board (i.MX6Quad) as shown in figure 1. The IR camera instead of RGB camera is applied to driver condition monitoring and drowsiness detection technology by extracting driver's facial feature information robustly against daytime, night-time and frequent change of brightness around face. The headphone type EEG sensor also is used to minimize the user's discomfort.

II. RELATED WORKS

Various studies have been conducted to detect driver drowsiness and driver fatigue. We believe this is the first work to extract features from EEG-signals and IR-images and apply them to the drowsiness estimation model. In this subsection, the related works are divided into two parts: 1) Research trends in image-based driver drowsiness and fatigue estimation. 2) Research trends in EEG-signals and sensor data convergence-based driver drowsiness and fatigue estimation.

A. Image-based Driver Drowsiness Detection

Most of the methods proposed for image-based drowsiness detection use features related to the condition of the eye. Detecting eyes that are closed or open is used in many methods. The simplest way to estimate drowsiness is to use the eye closure feature. Despite its simplicity, this method provides beneficial information about the driver's drowsiness. The eye closure feature can be used as two types of features for estimating driver drowsiness: (1) continuous eye closure and (2) percentage of eye closure for a specific time. In the continuous eye closure method, drowsiness is detected if the eyes are closed for a specified period. This method is not efficient and unreliable, as driver drowsiness is detected only when the driver's keep eyes closed for a specified period. This method is usually associated with other features to increase system accuracy. Another method is based on the percentage of eye closure (PERCLOSE). The PERCLOS method calculates the number of frames with no pupils detected and divides them by the total number of frames over a specific time interval. Therefore, this metric can only be used as a single value for a specified period [11]–[13]. The pupils and IR beams were detected considering the characteristics of the IR beam reflected from the pupil. The above features were used to identify closed or open eye conditions. Then PERCLOS was calculated and used for driver drowsiness detection. Several studies, eyelid distance, Eye Blink Speed, and Eye Blink Rate, were also used for drowsiness detection. In addition to these techniques, deep learning-based methods have been proposed. These methods proposed new architectures based on deep learning. State-of-the-art performance is continuously being updated using deep learning-based methods. However, this method's success requires a large amount of data for deep learning and requires high hardware performance.

B. Biometric-signal-based Driver Drowsiness Detection

The most commonly attempted physiological method for driver drowsiness is to obtain and utilize brain activity (EEG-

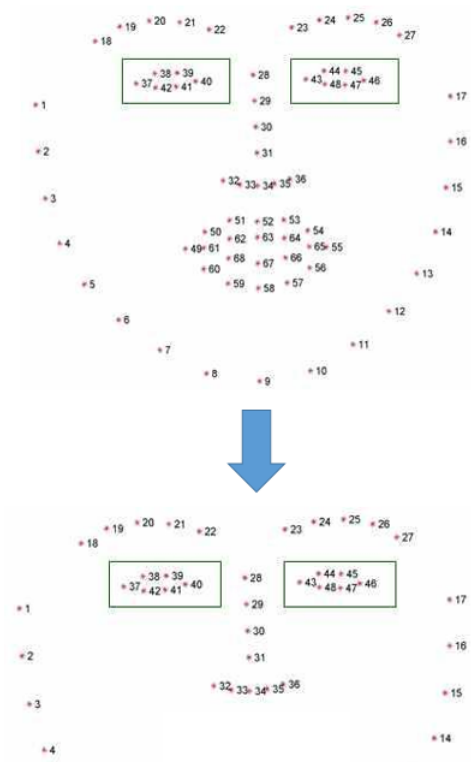


Fig. 2. Modified number of facial feature points

signals) information. Since EEG measures neuronal activity signals in the human brain, it is considered the most important and reliable way to detect brain activity signs when drowsiness occurs. Recently, EEG signals and blink signals are used for wheelchair control. These are used to construct the electrical voltage to control the wheelchair. The data values obtained from the EEG equipment are compared to a database stored on the computer. The signals are compared to database values stored on the computer and take specific actions assigned to them. Similarly, EEG-signals can also be used to detect the driver's drowsiness. The EEG sensor can be used to calculate the level of blinking in the eye and compare the stored EEG-signal values of the average driver's drowsiness.

III. METHOD

A. IR-image Based Model

In this model, the features related to drowsiness mainly appear in the area around the eyes, so we tried to extract the drowsiness features by considering only the area around the eyes instead of the entire face area [14], [15]. To reduce the computational power of the target board, we extract 39 landmarks (face feature points) from the face area instead of 68 [16], [17], as shown in figure 2.

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|} \quad (1)$$

where p_i is the landmark result as shown in figure 2. The eye area can be detected using the landmark result, and the

EAR (Eye Aspect Ratio) [18] is used to determine the degree of eye-opening and closing. As shown in figure 3, the EAR is the value of the ratio of the width and height of the eye using six eye-related feature points in the green box in figure 2. The smaller the EAR value, the lower the eye is opened. From these EAR values, it is possible to estimate how much the eye is opened. We estimate the degree of opening and closing of the eyes in three states: the EAR in the normal, the open EAR in the eyes open, and the closed EAR in the eyes closed. The average of the five most significant EAR values in a 1-minute video clip is the open EAR, and the average of the five smallest EAR values is closed EAR. In figure 3, the upper shows the waking state, and the bottom shows the sleepy state. In the waking state, the blinking of the eyes is relatively constant, and the proportion of frames that open the eyes is high. However, in the sleepy state, the proportion of frames that close the eyes is high.

1) *Feature Extraction From IR-images*: We tried to estimate the state of drowsiness by extracting drowsy features that are known to be useful, such as the degree of closing the eyes, the number of blinks, and irregular eye movements. The details of the drowsiness features are as follows: 1) Percentage of frames with eyes opened: Measures the rate of frames with 80% or more eyes open within a specified time. 2) Percentage of frames with eyes closed (PERCLOS): Measures the percentage of frames with 20% or fewer eyes open within a specified time. 3) Average eyes closed duration: Measures the average time during the eyes are closed when blinking. 4) Maximum eyes closed duration: The maximum time the eyes are closed when blinking. 5) Eyeblink frequency: how often the eye blinks within a specified period of time.

2) *Eye State (Opened/Closed) Estimation Method*: The proposed eye state (opened/closed) estimation method is as follows. We used the pattern of EAR values obtained in the awake state. It was assumed that the average of the five maximum EAR values in each video sequence was obtained and that the average of these values represents the EAR value when the experimenter's eyes are opened 100%. Similarly, the average of the five minimum EAR values was assumed to be the EAR value when the subject's eyes are opened 0% (i.e., eyes are closed 100%). The degree of eye closure can be estimated from the EAR value. The case where the eyes are opened more than 80% is called the eyes opened state, and the case where the eyes are closed more than 80% is called the eyes closed state. Besides, it was estimated as blink when the eyes closed state was continuous. Figure 4 shows the process of extracting samples and features from the IR video and determining whether they are drowsy.

B. EEG-signal Based Model

In this paper, we tried to use the detection of EEG changes in the frontal lobe as auxiliary information for estimating the driver's drowsiness. For the EEG measurement, EMOTIV EPOC+ was used as EEG measurement equipment. The EMOTIV EPOC+ is designed for scalable and contextual human brain research and provides access to professional grade brain

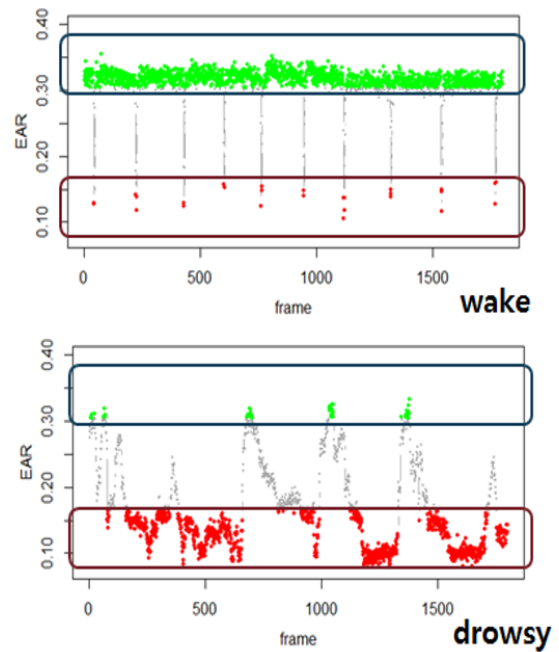


Fig. 3. Change of EAR value for each frame in the wake and drowsy. (green: eyes open, red: eyes closed)

data with a quick and easy to use design. The program to record data from EEG equipment used the API provided by EMOTIV. The EMOTIV API saved EEG data in CSV format as shown in figure 5. The exact start time is recorded in the file name. Since 128hz equipment is used, 128 rows are recorded in one second. In each row, voltage values from 14 channels, such as AF3 and F7, are recorded. Along with this, quality, which provides information on how reliable the measured values are, is also recorded.

1) *Feature Extraction From EEG-signals*: In this study, we acquired EEG data from AF3 and AF4 are used based on the 10-20 system of electrode placement, proposed by the International Federation of Societies for Electroencephalography and Clinical Neurophysiology in 1958, has been the international standard for recording routine scalp EEG for clinical use. The seven sample data were generated from 1 minute EEG data.

We used EEG-signal value and time information in synchronization with IR-image time information. Each sample was created by shifting the 30-second window at 5-second intervals. Figure 6. shows an example image of the result of synchronizing the blue line (EAR) and EEG-signal (red line) by IR image. The average and standard deviation of the top 5 maximum values among EEG-signals in the extracted 30-second intervals were obtained. Among the maximum values, values that are outside (average + 2 * standard deviation) are removed because they are outliers. The value between - 20% and + 20% in the average of the maximum EEG-signal values were defined as the eyes closed state. We extracted the drowsiness features of EEG-signals as follows: 1) The proportion of open eye frames. 2) The proportion of closed eye frames (PERCLOS). 3) Average eye blink duration (the length

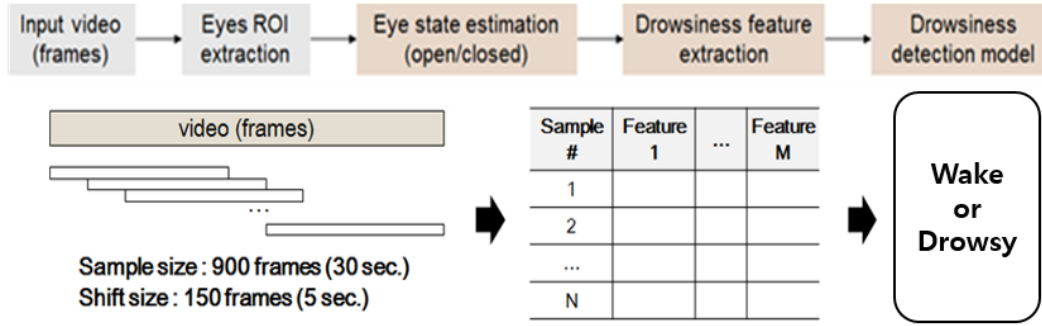


Fig. 4. Processing of extracting samples and features from the IR video.

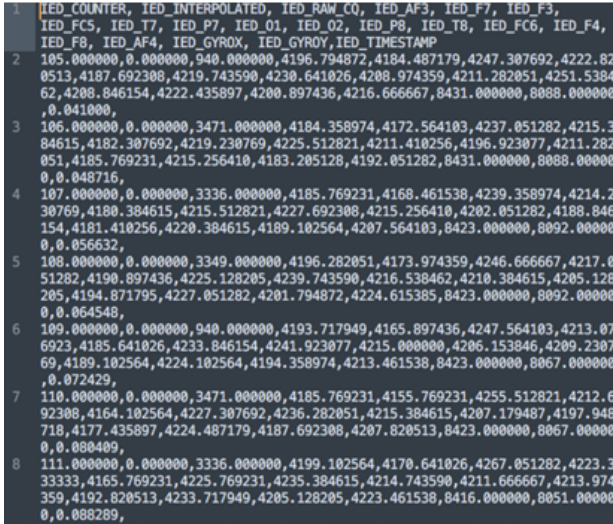


Fig. 5. Example of EEG-signals out from EMOTIV EPOC+

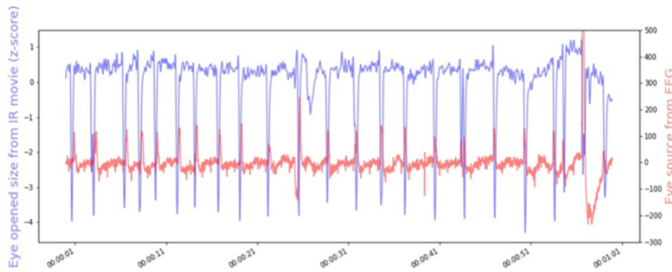


Fig. 6. An example image of the result of synchronizing the blue line (EAR) and EEG-signal (red line) by IR image.

of time the eyes are closed in one blink event). 4) Maximum eye blink duration (the length of time the eyes are closed in one blink event). 5) Eye-blink frequency.

IV. IR+EEG INTEGRATED DATA-BASED MODEL

In this model, IR-images and EEG-signals-based data-sets were constructed, and time synchronization was performed to integrate the two data-sets. Five feature values were extracted

from IR-images and EEG-signals, respectively. Each sample was built with ten combined feature vector data and used as input for machine learning to determine whether the driver is drowsy or not.

V. EXPERIMENTS

The output of the driver drowsy prediction model is wake or drowsy and the evaluation indicators are accuracy, sensitivity, and specificity. The sensitivity of a test is also called the true positive rate (TPR) and is the proportion of samples that are genuinely positive that give a positive result using the test in question. The specificity of a test, also referred to as the true negative rate (TNR), is the proportion of samples that test negative using the test in question that are genuinely negative.

A. Training Details

The training and test data-set were acquired by classifying between wake and drowsy cases. Six men and four women in their 20s participated in the collection of EEG and IR image data. The EEG and image data are time-synchronized, the wake and drowsy states are composed of half and half, and a total of 600 samples. Of the 600 data obtained, 360 samples participated in the training, and the remaining 240 samples involved in the test. For retraining the facial feature point detector, we made a 3,571 face landmark dataset from the infrared camera since we could not find IR camera-based face landmark datasets. To detect the face landmarks, We used a method of regressing Local Binary Feature (LBF), which consists of local binary feature mapping and global linear regression using random forests. Four drowsiness estimation models based on machine learning were implemented, and performance was compared and analyzed. The machine learning methods used are linear regression (LR), k-nearest neighbor (KNN), support vector machine (SVM), and random forest (RF).

B. Experimental Results

Among the threshold-based drowsiness estimation performance widely used, the results of the drowsiness detection model that is extracted and applied as a feature of the eye closed state are as follows. The method using average and

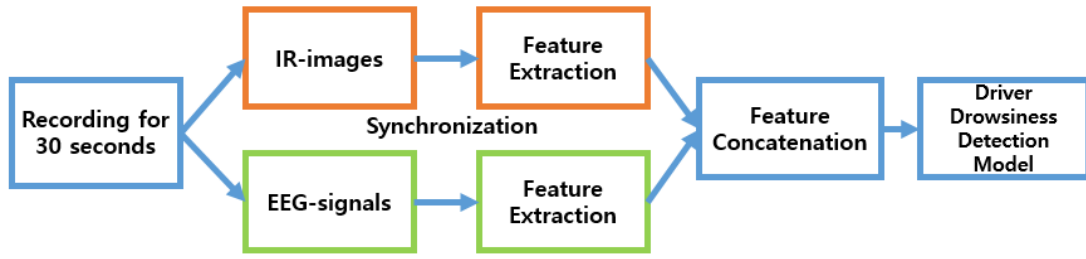


Fig. 7. Work-flow for driver drowsiness prediction model using concatenated features extracted from IR-images and EEG-signals.



Fig. 8. EMOTIV EPOC+ EEG equipment, an example of wearing and 10-20 system of electrode placement.

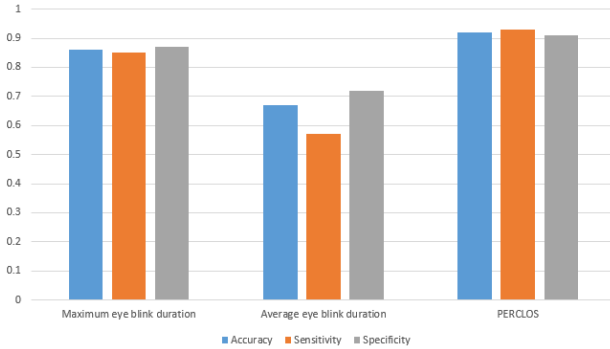


Fig. 9. Comparison of threshold-based drowsy detection method performance.

maximum eye blink duration was 62%, and 86% accuracy, respectively, and the method for detecting drowsiness using PERCLOS was 92% accuracy. The method using PERCLOS performed better than the others. We adopted the 5-fold cross-validation method and constructed training and validation data using the 360 samples mentioned above. We applied the combined feature vectors were extracted from IR-images and EEG-signals to LR, KNN, SVM, and RF machine learning methods for training the drowsiness detection model.

The LR and KNN methods performed relatively poorly, and the RF method showed 98% accuracy and the highest performance. After selecting some of the drowsiness features which we proposed, we trained it with RF classifier since it shows highest performance and compared the drowsiness detection performance. As a result, we found that the more features, the higher the drowsiness detection accuracy. We tested our drowsiness detection algorithm on the targetboard using

TABLE I
RESULTS OF 5-CROSS VALIDATION OF LR, KNN, SVM, AND RF
MACHINE LEARNING METHODS .

Method	Accuracy	Sensitivity	Specificity
LR	0.92	0.94	0.90
KNN	0.95	0.94	0.97
SVM	0.95	0.93	0.98
RF	0.98	0.97	0.98

1GHz i.MX6Quad and 1GB RAM. On the target board, the processing time per image frame is about 60ms. The optimal parameters were learned through 5-fold cross-validation, and the highest performance RF model was applied to the test dataset, and the results are shown in Table 2. As shown in Table 3, the performance of the model using IR-images and EEG-signals together shows improved drowsiness detection sensitivity than the model using only IR-images. The reason for the improved sensitivity is that the eyes are not closed in the image, but the EAR is measured at a small value. However, the EEG-signals did not show drowsiness, so it improved performance.

TABLE II
DROWSINESS DETECTION PERFORMANCE OF RF MODEL ACCORDING TO
FEATURE SELECTION.

Features	Accuracy	Sensitivity	Specificity
PERCLOS	0.92	0.93	0.91
PERCLOS + Percentage of frames with eyes opened	0.94	0.94	0.93
PERCLOS + Maximum eyes closed duration	0.95	0.92	0.98
Average eyes closed duration + Maximum eyes closed duration	0.95	0.92	0.97
All	0.98	0.97	0.98

VI. CONCLUSION

This paper extracts features through the merging of IR-images and EEG-signals and implements a low-cost real-time driver condition monitoring system using machine learning. The performance of the IR-images and EEG-signals fusion-based model is about 0.01 and 0.02 higher in accuracy and sensitivity than the IR-images-based model. The drowsiness

TABLE III
PERFORMANCE EVALUATION AND COMPARISON OF DROWSINESS
DETECTION MODEL.

	IR-images	EEG-signals	IR-images + EEG-signals
Accuracy	0.85	0.82	0.86
Sensitivity	0.87	0.86	0.89
Specificity	0.83	0.80	0.83

detection model based on the merging of IR-images and EEG-signals for driver drowsiness is robust and shows a performance improvement, unlike the single information-based model. We tested our drowsiness detection algorithm on the target board using 1GHz i.MX6Quad and 1GB RAM. On the target board, the processing time per image frame is about 60ms, so that it can process about 17 frames per second. This processing time is suitable for the driver's drowsiness detection systems.

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